



**Question 1** (10 marks)

(a) Explain the importance of glycine in the formation of collagen triplex helix. [2max]

1. Glycine is small/smallest amino acid; (*Bonus*: for identifying R as Hydrogen)
2. Allows formation of **kinked** helix;
3. Allows **tight fitting** of the 3 helices/ triple helix/tropocollagen/does not disrupt triple helix;
4. By decreasing steric hindrance;
5. Allows formation of **interchain hydrogen bonds** are formed;
6. between the peptide -NH group of the glycine residues and the peptide -CO groups of residues on the neighbouring chains;

® glycine is hydrophobic and this makes collagen insoluble – as question asks for the formation as in structure.

NB: collagen triple helix = tropocollagen = basic unit/ building blocks of collagen

(b) Describe how association between triple helices allows the formation of fibers with high tensile strength. [3 max]

1. Each tropocollagen interacts with other tropocollagen **running parallel** to it;
  2. **Covalent bonds** formed;
  3. between the side-chains of the **lysine residues** between tropocollagen;
  4. **Staggered arrangement** of tropocollagen in parallel chains;
  - ® ends of tropocollagen is staggered.
  5. No weak areas/ more difficult to break;
  6. Cross-links hold many tropocollagen side by side, forming a collagen fibril;;
  7. Collagen fibrils further aggregated to form collagen fibres;
- Allows the formation of a very stable structure with no capacity to stretch;

(c) Distinguish between gene mutations and RNA splicing mutations. [2]

| <i>Gene mutations</i>                                                           | <i>RNA splicing mutations</i>                     |
|---------------------------------------------------------------------------------|---------------------------------------------------|
| involve changes at the <b>DNA</b> level;                                        | involve changes at the <b>RNA</b> level;          |
| involve insertion, deletion, substitution of base sequences; (mention of any 2) | involve the joining of exons/ removal of introns; |
| Genetic mutations occur in prokaryotes and eukaryotes;                          | RNA splicing mutations occurs in eukaryotes only  |

Gene mutations are inheritable but not RNA splicing mutations. (NB: may not be true as the RNA splicing mutations involved proteins which are coded for by DNA.) (1/2 m for complete comparison)

(d) Explain why gene mutations do not always produce mutated collagen whereas mutations involving RNA splicing will always produce mutated collagen. [3]

1. Gene mutations that involve **substitution**;
2. May result in the **same amino acid** being coded for;
3. **Degenerate** code/ same amino acid being coded for by different codons/ a specific example;  
     ® degenerative code

| An example of RNA splicing mutation<br>(any one -1 mark each) | Effect (i.e. production of mutated collagen)<br>(any one -1/2 mark each)                   |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| different combinations of <b>exons</b> being produced         | Different primary sequences of amino acids resulting in different protein(mutated protein) |
| an exon is lost/ wrong excision of exons                      | Large number of bases and hence amino acids is lost/ as above;                             |
| <b>Introns</b> not removed                                    | Introns translated and additional amino acids change the protein structure;                |

### Question 2 [10 marks]

- (a) Gene – sequence of bases;  
 - that codes for a polypeptide/product;  
 Alleles – alternative forms of a gene / Have slightly different base sequences;  
 - Occupies the same gene loci;  
 - codes for different products; (any 2 points)
- (b) Let A = allele that codes for enzyme 1 and a= allele that does not code for enzyme ;  
 Let B = allele that codes for enzyme 2 and b = allele that does not code for enzyme 2;  
 And A and B are completely dominant over a and b respectively;

Parents  $AaBb$   $\times$   $AaBb$  1m correct genotypes

Gametes  $\downarrow$  meiosis  $\downarrow$  meiosis

$AB$   $Ab$   $aB$   $ab$   $AB$   $Ab$   $aB$   $ab$  1m correct gametes (all or none)

| gametes | AB   | Ab   | aB   | ab   |
|---------|------|------|------|------|
| AB      | AABB | AABb | AaBB | AaBb |
| Ab      | AABb | AAbb | AaBb | Aabb |
| aB      | AaBB | AaBb | aaBB | aaBb |
| Ab      | AaBb | Aabb | aaBb | aabb |

1 m for punnet square

Genotypes: A\_B\_ = Cyanogenic = 9/16;  
A-bb = acyanogenic  
aaB\_ = acyanogenic  
aabb = acyanogenic } = 7/16;

1 m for correct genotypes and phenotypes

(4 max.)

- (c)
- (i) – below  $-4^{\circ}\text{C}$ : all plants are acyanogenic;
    - As cyanogenic plants' cell membrane breaks;
    - releasing cyanide which will kill itself;
    - Slugs not active and hence predation not a problem;
    - Hence selective advantage is not to produce cyanide;
  - above  $10^{\circ}\text{C}$ : mostly cyanogenic;
  - slugs present as selective pressure;
  - cyanide production kills slugs;

- these plants survived, reproduce and pass on desirable traits to offspring; (4 max.)

### Question 3 [10 marks]

(a) State the precise location of the four stages. [2]

| Respiratory Stage         | Location in cell                                            |
|---------------------------|-------------------------------------------------------------|
| Glycolysis                | Cytoplasm/ cytosol;                                         |
| Link reaction             | Mitochondrial matrix                                        |
| Krebs cycle               | Mitochondrial matrix <i>R: Mitochondrion / matrix alone</i> |
| Oxidative phosphorylation | Inner mitochondrial membrane / cristae / stalked particles; |

(b) Explain the results of the graph

(i) Between points A and B [2]

- 1 unable to make ATP;  
/ ADP limiting / no ADP added;
- 2 oxidative phosphorylation/aerobic respiration, unable to take place;
- 3 no O<sub>2</sub> taken up / no change in O<sub>2</sub> concentration / remains constant;
- 4 ref. to glycolysis can proceed still to convert glucose to pyruvate;

(ii) Between points B and C [2]

- 1 ATP formed / by chemiosmosis / by using energy built up from ETC;
- 2 O<sub>2</sub> level decreases;
- 3 aerobic respiration / oxidative phosphorylation, proceeds;
- 4 but slows down as ADP is used up;

(iii) After point D [2]

- 1 aerobic respiration/oxidative phosphorylation, not taking place;
- 2 glucose / substrate used up;

- 3 oxygen level remains constant/same, does not decrease further;
  - 4 ADP will not be used up (for chemiosmosis);
- (c) Radioactive amino acids taken up by ribosomes of RER for synthesis of polypeptide; Radioactive highest 1 min after introduction at 120; Polypeptides entered into lumen of RER for modification; radioactivity at 68 20 mins after introduction; Packaged to be transported to Golgi for further modification to become functional; Q data; Packaged into vesicles and transported away from Golgi; Q data.

#### Question 4 (10 marks)

- (a) Describe how enzymes are used to construct the DNA fragment shown above.  
[2]

- **Same** restriction enzyme is used to isolate all the DNA sequences (A to D);
- So that *complementary sticky ends* will be produced;
- **Hydrogen bond** formation will occur between these sticky ends;
- **DNA ligase** and ATP added to form **phosphodiester** bonds between these fragments;

OR

- Restriction enzyme that creates blunt ends is used to isolate these DNA sequences;
- Add linkers to these blunt ends to make them sticky;
- **Hydrogen bond** formation will occur between these sticky ends;
- **DNA ligase** and ATP added to form **phosphodiester** bonds between these fragments;

NB: These DNA fragments cannot be added all together as different combinations of the fragments will be formed.

- (b) Suggest why the following sequences were included in the DNA introduced into the clover.

- (i) Sequence B [2 max]

- Desired protein is expressed only in photosynthesizing cells/ leaves which the sheep feed on;;
- Enzyme RuBP is used in the dark reaction of photosynthesis;
- Enhancer sequence will increase the rate of transcription/ enhancer will increase expression of this enzyme;
- i.e. increase in photosynthesis and hence increase expression of this gene;

- (ii) Sequence D [2 max]
- Protein taken in by the sheep will be broken down/digested in the rumen and some may be excreted;;
  - Allows the incorporation of this protein into wool, leading to increase yield of wool;;
- (c) The Ti plasmid in *Agrobacterium* is often used in the genetic engineering of plants. Describe how the Ti plasmid can be used as a vector to introduce the above length of DNA into clover. [2]
- Tumor causing genes of T-DNA is removed;
  - Cut the T-DNA and Gene of interest (GOI) with the same restriction enzyme;
  - Gene of interest inserted into **T-DNA**;
  - Recombinant plasmid introduced into host cells;
  - Integration of the T-DNA (with GOI) into plant's genome;
- (d) Briefly describe two methods which can be used to transfer the recombinant Ti plasmid into *Agrobacterium*. [2]
- (1) Calcium chloride and heat shock /*Electroporation*;  
Open up pores in the cell membrane to facilitate uptake of recombinant plasmids /Use of *electric currents* to create transient pores in the membranes to facilitate uptake;
  - (2) Gene Gun;  
Recombinant plasmid is coated onto gold 1 micron particles and shot into the bacterial cells;

### Question 5(a) (Essay)

(a) Using named examples, compare and contrast between storage and structural forms of carbohydrate. [8]

|                         |                                                                                                                     |                                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Comparison features     | Storage : Starch                                                                                                    | Structural: Cellulose                                                          |
| Monomers                | $\alpha$ - glucose                                                                                                  | $\beta$ -glucose                                                               |
| Bond                    | $\alpha$ - 1,4 glycosidic bond                                                                                      | $\beta$ -1,4 glycosidic bond                                                   |
| Orientation of monomers | All oriented in the same direction                                                                                  | Alternate residue rotate by $180^\circ$                                        |
| Shape                   | Helix                                                                                                               | Straight parallel chain                                                        |
| OH group                | Point inwards of helix                                                                                              | Point upwards and downwards of each chain                                      |
| Properties              | Compact because can pack many monomers (a lot of monomers per unit volume) due to helical                           | High tensile strength due to hydrogen bonds between parallel chains            |
| Properties              | Insoluble so that will not dissolve away due to its large size/ OH groups not available to hydrogen bond with water | Partially soluble as some OH groups are free to form hydrogen bonds with water |
| Properties              | Does not affect the osmotic pressure as the concentration of the cell is not affected.                              | Easily hydrolysable as water molecules are available for hydrolysis            |

|            |                                                                               |                                                                                                                                                                                      |
|------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Properties | Lots of energy is stored due to large number glucose released upon hydrolysis | Permeable to all substances- as OH groups are free to form hydrogen bonds with water, water channels and gaps in cellulose structure allow substances to move within and through it. |
|------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Question 5(b)

| Features                         | Prokaryote                                                                                                                  | Eukaryote                                                                                                                                                                                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control at chromosomal level     | - No control at chromosomal level                                                                                           | - Controls at chromosomal level                                                                                                                                                                                                                |
|                                  | - DNA not associated with histones, hence no histone modification possible                                                  | - Involved histone acetylation and chromatin remodeling                                                                                                                                                                                        |
| Control at transcriptional level | - Operator involved in control of transcription; operator is proximal/near to the genes under its control                   | - Involved control elements; that may be proximal/near (e.g. upstream regulatory elements) or distal/further (e.g. enhancers and silencers)<br><br>Extra:<br><br>- Further involved DNA methylation that reinforced transcriptional repression |
|                                  | - Some operons (e.g. lac operon) underwent both positive (catabolite activator protein) and negative regulation (repressor) | - Involved specific transcription factors (e.g. activators and repressors); combinatorial control and coordinated control involved<br><br>OR<br><br>- Control elements and specific transcription factors increased (activated) or decreased   |

|                                       |                                                                                           |                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                       |                                                                                           | (repressed) the rate of transcription of the target gene                                                                                    |
| Type of mRNA produced                 | - Polycistronic mRNAs                                                                     | - Monocistronic mRNAs                                                                                                                       |
|                                       | - Operon system: multiple related structural genes controlled by the same promoter        | - Each gene controlled by its own promoter                                                                                                  |
| Control at post-transcriptional level | - Transcription and translation occur simultaneously due to absence of a nuclear membrane | - Nuclear membrane separates the processes of transcription and translation                                                                 |
|                                       | - No post transcriptional modifications occur                                             | - Post transcriptional modifications occur. Eg<br>-Alternative RNA splicing, - Addition of 5' cap, 3' poly(A) tail) occurred<br><br>Any one |

### Question 5c

- Sequence of bases in gene/on DNA determine amino acid sequence;
- Transcription to form mRNA;
- Codons on mRNA;

- Reference to base-pairing;
- A to U, T to A and C to G;
- Carries the information for the protein to be synthesized;
- Shine-Dalgarno sequence binds to small subunit of ribosome for protein synthesis initiation;
- Specific tRNA for each amino acid;
- Has anticodon to bind to mRNA codon;
- 3' end has an attachment site for amino acid;
- To bring the appropriate amino acid to the mRNA for protein synthesis;
- Complex of rRNAs and ribosomal proteins;
- Form the large and small subunits of ribosomes;
- Role of A and P sites on ribosomes in translation;

### Question 6(a)

(a) Compare and contrast between cyclic and non-cyclic photophosphorylation pathways in photosynthesis. [4]

| Comparison features                            | Non- cyclic                                                                                                                                                                                  | Cyclic                                                                                                                                                               |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pathway of electron                            | Non-cyclic                                                                                                                                                                                   | Cyclic                                                                                                                                                               |
| Photosystems involved                          | PS 1 and PS 2                                                                                                                                                                                | PS 1                                                                                                                                                                 |
| First electron donor (source of electron)      | Water                                                                                                                                                                                        | PS 1                                                                                                                                                                 |
| Last electron acceptor                         | NADP                                                                                                                                                                                         | PS 1                                                                                                                                                                 |
| Products                                       | ATP, NADPH, oxygen                                                                                                                                                                           | ATP                                                                                                                                                                  |
| Establishing proton gradient for ATP synthesis | High [H <sup>+</sup> ] in the thylakoid space is due to photolysis of water and active transport of H <sup>+</sup> from the stroma, across the thylakoid membrane, into the thylakoid space. | High [H <sup>+</sup> ] in the thylakoid space is due to active transport of H <sup>+</sup> from the stroma, across the thylakoid membrane, into the thylakoid space. |

### Question 6 (b)

Distinguish between gene and chromosomal aberration. [6]

| <b>Features</b> | <b>Gene Mutation</b>                                                                                                                                          | <b>Chromosome Mutation/ Chromosome Aberration</b>                                                                                                                                                                                                                                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition      | Change in the structure of the DNA/ the <b>base sequence</b> / nucleotide sequence;<br>Occurs at a single locus on a chromosome;                              | Chromosomal aberration or mutation is defined as a <b>change in the structure</b> of a chromosome;<br>or the <b>number of chromosomes</b> ;                                                                                                                                                                                                             |
| Mechanism       | Brought about by deletion, insertion, substitution or inversion of bases;                                                                                     | Brought about by:<br>1. Deletion, inversion, translocation or duplication of the chromosome;<br>2. Non-disjunction (of homologous chromosomes or sister chromatids during mitosis or meiosis);                                                                                                                                                          |
| Results         | Only 1 gene is affected;                                                                                                                                      | Effect is more profound;<br>as several genes are affected;                                                                                                                                                                                                                                                                                              |
|                 | Base substitution may lead to the formation of new alleles;                                                                                                   | Base sequence of individual gene usually not affected;<br><br>Affects the:<br>1. Positions of several genes on a chromosome;<br>2. The number of genes on a chromosome;<br>3. Number of chromosomes in a cell;<br>(max of 1 mark for this point)<br>a) Loss or gain of 1 chromosome/ aneuploidy;<br>b) Loss or gain of sets of chromosomes/ polyploidy; |
| Occurrence      | Occur anytime;<br>(NB: For gene mutation as well as chromosomal aberration in chromosome structure, these are random events and therefore can occur anytime.) | <i>For chromosome aberration in structure:</i><br>Occur anytime;<br><i>For chromosome aberration in number:</i><br>Occurs when homologous chromosomes or sister chromatids fail to separate;<br>During mitosis and/or meiosis;                                                                                                                          |
| Frequency       | More frequent as there are many more genes than chromosomes;<br>(NB: it is necessary to qualify why the frequency of gene mutations appear to be              | Less frequent;                                                                                                                                                                                                                                                                                                                                          |

|                         |                                                                                                                                                                                                                                                                                    |                                                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|                         | higher)                                                                                                                                                                                                                                                                            |                                                                                                                               |
| Evolutionary importance | Involves introduction of new alleles into the gene pool;                                                                                                                                                                                                                           | Involves reshuffling of existing alleles in the gene pool;                                                                    |
| Consequence             | <p><i>Can have different consequences ranging from no effect to fatal.</i></p> <p><b>May</b> have <u>no effect</u> (degenerate code);<br/> <b>May</b> be <u>fatal</u> eg. frame-shift mutations;<br/> (NB: Not all gene mutations are less severe than chromosomal mutations.)</p> | Generally will cause the expression of a <u>different phenotype</u> (® genotype) eg. mental retardation as in Down's syndrome |

### Question 6c

1. Stem cells are capable of dividing by mitosis;
  2. Capable of making identical copies of themselves for the life time of the organism;
  3. Self-renewal;
  4. Resulting cells continue to be unspecialized;
  5. Divide to give rise to 2 equally sized cells;
  6. For further division into stem cells;
  7. Divide to give rise to 2 unequal cells;
  8. 1 cell which retains the ability to divide while the other becomes specialised;
  9. Stem cells are unspecialized;
  10. Do not have any tissue specific structures that allow it to perform specialized functions;
  11. Undergo differentiation;
  12. To repair/ replaced damaged/ diseased cells;
  13. Adult stem cells;
  14. Exhibit ability to form specialized cell types of other tissues;
  15. Transdifferentiation/ plasticity;
- [6 max.]

Any two of the following examples [4 max.]

16. Embryonic stem cells;
17. Derived from the inner cell mass of blastocyst;
18. Pluripotent;
19. Can reproduce indefinitely;
20. Gives rise to ectoderm, mesoderm and endoderm;

- 21. Hematopoietic stem cells;
- 22. Found in bone marrow;
- 23. Multipotent;
- 24. Gives rise to red blood cells, white blood cells, macrophages and platelets;
  
- 25. Skeletal muscle stem cells;
- 26. Also known as satellite cells;
- 27. Triggered to proliferate as a result of injury/ weight bearing/ exercise;
- 28. Unipotent;
- 29. Gives rise to myogenic precursor cells;
- 30. Differentiate onto myofibrils.